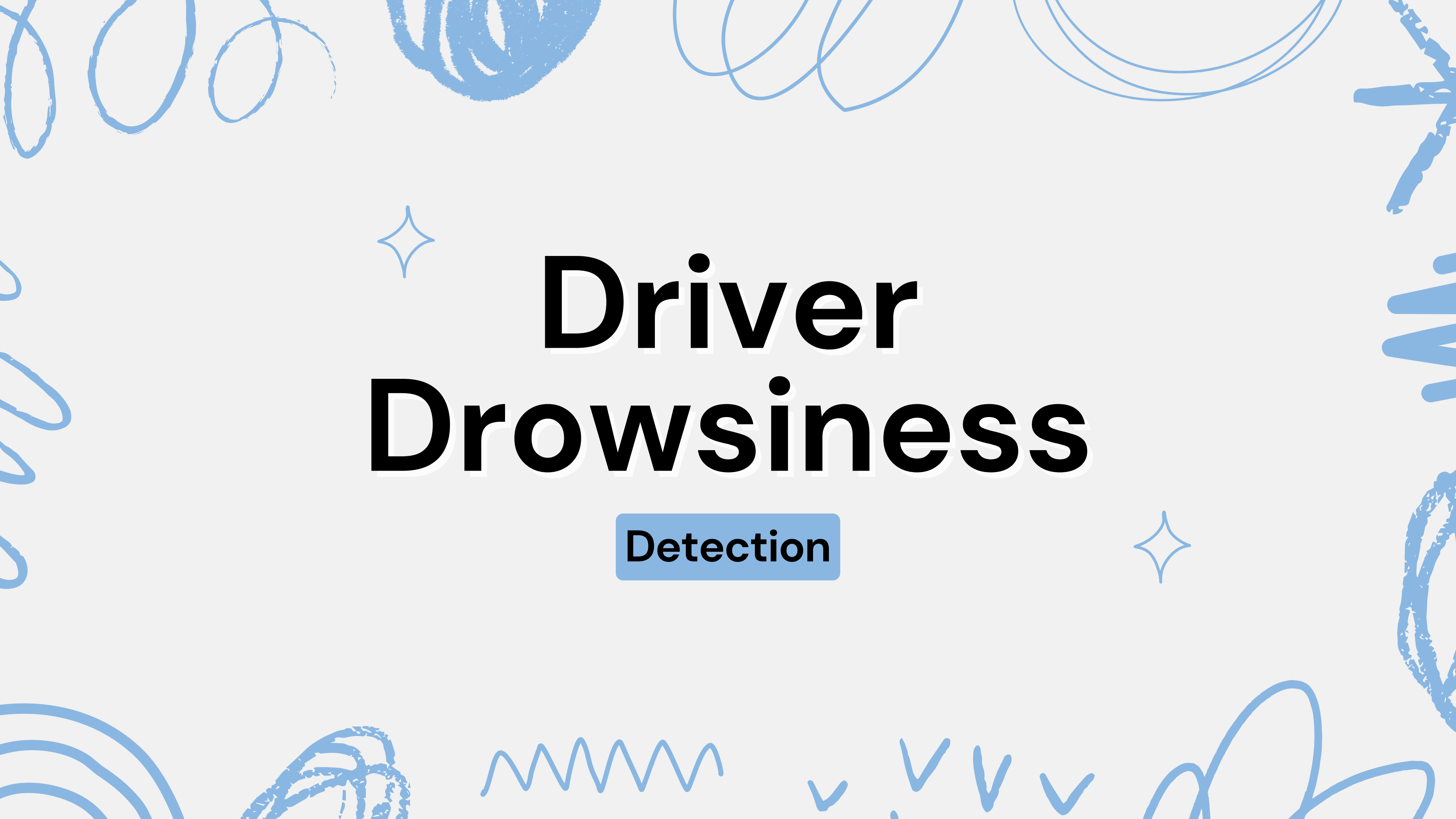




Driver Drowsiness

Detection

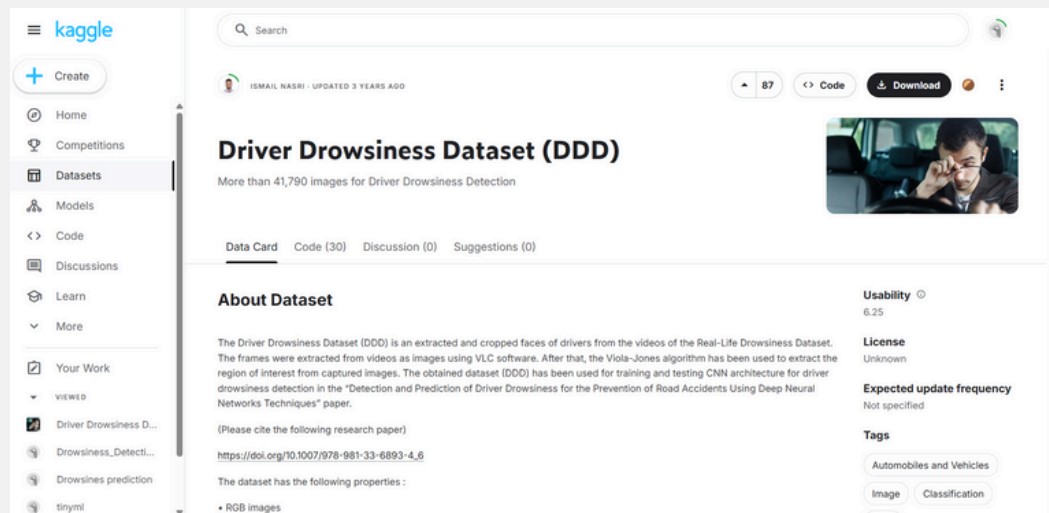
Problem & Motivation

Drowsy driving leads to delayed reaction time and increased accidents. Our goal is to detect drowsiness in real-time using deep learning and computer vision techniques.



Dataset & Preprocessing

Used the Driver Drowsiness Detection Dataset (Kaggle).
Balanced classes with thousands of images each. Applied transformations like rotation, flipping, color jitter, Gaussian blur, and normalization to improve model robustness.





Custom CNN



MobileNetV3



CNN
+
MobileNetV3



Model Develop

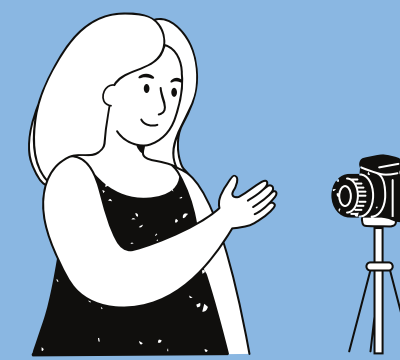
Exploring model



ResNet50



CNN
+
ResNet50



YOLO

Detection Methods

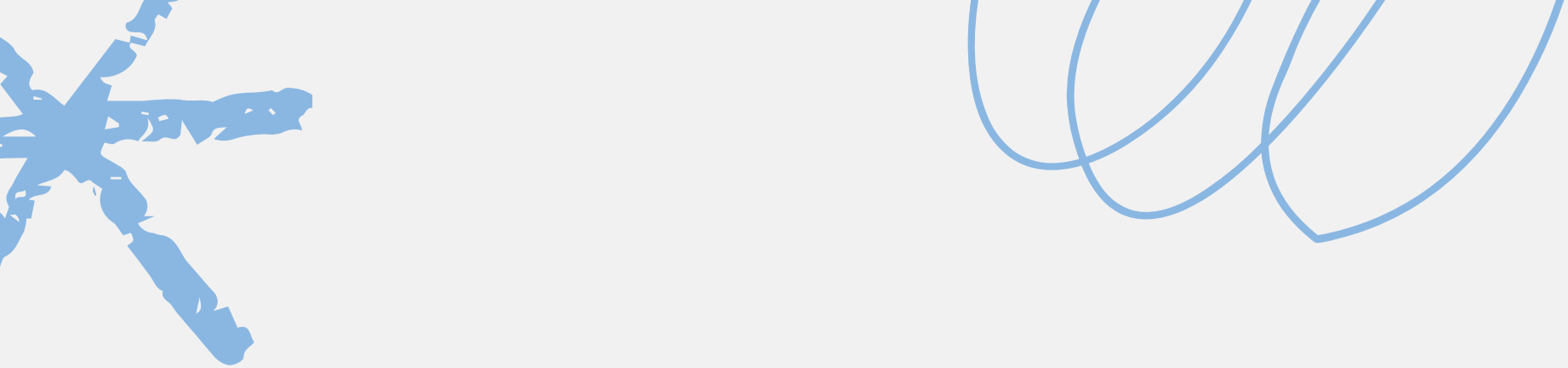
Haarcascade for
single eye

Haarcascade for
both eye

Haarcascade for
face

YOLOv8n





Technique to solve the problem

01.

Increase Learning
Rate and Epoch

02.

Early Stop

03.

Drop out



Model Comparison

01

Custom CNN

Accuracy: ~78%

F1-Score: ~0.78

02

ResNet50

Accuracy: ~95%

F1-Score: ~0.93

03

MobileNetV3

Accuracy: ~96%

F1-Score: ~0.95

04

CNN + ResNet50

Accuracy: ~97%

F1-Score: ~0.96

99.81%

CNN + MobileNetV3 Accuracy

With the model that combine the Custom CNN with the MobileNetV3 model can achieved best result with accuracy is 99.81% and F1-score is 0.998. For the average of both average and F1-score is ~99% and 0.98



Live Demo

Future Work



Future Works



01. Retrain YOLO for face detection

02. Track eye closure and yawning

03. Improve system UX/UI and deploy



Thank you very much!

Q&A Session